

13. Write a Pandas program to detect missing values of a given DataFrame and display True or False.

Aim

To write a Pandas program to detect missing values in a given DataFrame and display the result as True or False.

DATASET:

Name	Age	Marks	City
Anu	20	85	Chennai
Bala	21	Missing	Madurai
Cathy	Missing	90	Coimbatore
David	22	78	Missing
Esha	20	Missing	Chennai

```
import pandas as pd
# Read the CSV file
df = pd.read_csv(r"C:\Users\DELL\Documents\dataset for 13.txt")
print("Original DataFrame:")
print(df)
print("\nMissing Values (True/False):")
print(df.isnull())
```

OUTPUT:

```
Original DataFrame:
   Name  Age  Marks  City
0  Anu  20.0   85.0  Chennai
1  Bala  21.0    NaN  Madurai
2  Cathy  NaN   90.0  Coimbatore
3  David  22.0   78.0    NaN
```

	Name	Age	Marks	City
0	False	False	False	False
1	False	False	True	False
2	False	True	False	False
3	False	False	False	True
4	False	False	True	False

Result

The Pandas program successfully detects missing values in the DataFrame and displays True for missing values and False for non-missing values.

14. Write a Pandas program to find and replace the missing values in a given DataFrame which do not have any valuable information.

Aim

To write a Pandas program to find and replace missing values in a given DataFrame with appropriate values.

Dataset:

Name	Age	Marks	City
Anu	20	85	Chennai
Bala	21	Missing	Madurai
Cathy	Missing	90	Coimbatore
David	22	78	Missing
Esha	20	Missing	Chennai

Python code:

```

1 import pandas as pd
2 # Read the CSV file
3 df = pd.read_csv(r"C:\Users\DELL\Documents\dataset for 13.txt")
4 print("Before replacing missing values:")
5 print(df)
6 # Replace missing Age with mean Age
7 df["Age"] = df["Age"].fillna(df["Age"].mean())
8 # Replace missing Marks with mean Marks
9 df["Marks"] = df["Marks"].fillna(df["Marks"].mean())
10 # Replace missing City with "Unknown"
11 df["City"] = df["City"].fillna("Unknown")
12 print("\nAfter replacing missing values:")
13 print(df)

```

Output:

	Name	Age	Marks	City
0	Anu	20.00	85.000000	Chennai
1	Bala	21.00	84.333333	Madurai
2	Cathy	20.75	90.000000	Coimbatore
3	David	22.00	78.000000	Unknown
4	Esha	20.00	84.333333	Chennai

Result

The Pandas program successfully finds and replaces the missing values. Numerical missing values are replaced with their column mean, while the missing city value is replaced with "Unknown".